News Shocks in Prediction Markets

Real results, 2026-10-14

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The question driving this thesis

Do news shocks about a company produce a measurable, predictable price reaction in that company's Polymarket contracts?

Everything that follows answers this question directly.

Context

- Data: 6,928 resolved contracts across 86 companies (Apr 2024-Aug 2026), matched against 546,657 GDELT articles
- No existing tool answers the question above directly, so this project builds two new methods
- GDELT only timestamps articles to the day, so every result below measures the 24-hour price reaction, the only window this feed supports

Method: the mechanics

Matching picks the right article for a contract, combining text similarity and trading-volume spikes:

match_quality = w1 · logit(cosine similarity) + w2 · z(volume spike), then sigmoid

Consistency forces a company's related contracts (above $150, $160, $170) to react in the same direction, weighted by volume, via isotonic regression (PAVA).

Method: why these, and why they should work

Both methods are new combinations of established tools, not inventions from scratch.

Combining independent signals in log-odds space is standard practice (the same math behind naive Bayes classifiers). Volume spikes as a marker of real news arriving is a standard idea in market microstructure research, going back to classic event-study methodology. Isotonic regression is a long-established tool for order-restricted inference (Barlow, Bartholomew, Bremner & Brunk, 1972), and the logic that related contracts should price consistently is the same logic behind no-arbitrage pricing.

What's new: applying both combinations to this specific problem, matching news to prediction-market contracts and enforcing no-arbitrage-style consistency across a strike ladder.

Result 1: contracts do react to news, with one caveat found since

Method	Spearman ρ vs. actual price move	n
Matching score (text + volume)	**0.509**	14,717
Text similarity alone	0.233	14,717

This is half the answer to our question: a company's news and its market's trading behavior are measurably linked. Every later result depends on this step, it is how we built a trustworthy dataset of article-reaction pairs at all.

New finding (next section): part of the 0.509 number is inflated. Validated against the real stock's own reaction instead of Polymarket's own data, the matching score drops to ρ=0.029; text similarity alone holds up better (ρ=0.092). The volume component likely rode a coincidence specific to Polymarket's own data.

Result 2: the consistency correction makes predictions worse

Category	R^2, raw reaction	R^2, after correction
Price-target	0.084	0.073
Valuation	0.746	0.363

We expected this to help: related contracts should price consistently, by no-arbitrage logic. The correction smooths away contract-specific variation the prediction models depend on, so we predict from the raw reaction.

Result 3: models land close to guessing the average

Category	Best model	R^2	n (test)
All (pooled)	Linear	−0.010	5,151
Corporate event	Linear	−0.021	3,157
Price-target	LSTM	**+0.010**	598
Revenue-target	TCN	−0.016	9
Valuation	Linear only	−0.255	small

This is the other half of the answer. If Polymarket prices already reflect public news quickly, this is exactly the pattern we'd expect: a reaction (Result 1) whose size depends on information beyond the article itself.

Result 4: one of four pre-registered predictions holds up

Prediction	Tests whether…	Result
Richer embeddings beat simpler ones	text representation matters	Not supported
Best model differs by category	architecture choice matters	Not supported
Quantitative articles beat high-attention ones	article type matters	Supported (n=28)
Best ladder category beats corporate events	ladder structure adds signal	Not supported

Tested with bootstrap resampling clustered by real-world event, so duplicate coverage of one story isn't counted as independent evidence.

Result 5: Polymarket tracks the real stock's direction

Test	Spearman ρ	95% CI (company-clustered)
Direction vs. direction	**0.119**	[0.047, 0.188]
Size vs. size	0.015	includes zero

A company can show a huge real-world move alongside a tiny Polymarket reaction, or the reverse.

Result 6: a tradeable signal, for price-target contracts

```
reaction = efficiency_ratio &middot; theoretical_sensitivity &middot;
(stock_return / stock_volatility) + baseline_drift
```

Term	Value	What it is
theoretical_sensitivity	1.702	frictionless digital-option slope (Birnbaum 1968), not fit to our data
efficiency_ratio	0.093	share of that sensitivity these contracts realize
baseline_drift	−0.184	24h drift with no stock move (favorite-longshot correction, Snowberg & Wolfers 2010)

These contracts move at about 9% of the rate a frictionless market would.

Result 6: validated and backtested

Check	Result
Correlation across train / val / test	0.18 / 0.14 / 0.18
Company-clustered bootstrap	95% CI [0.043, 0.288]
Backtest vs. 5,000 random permutations	100th percentile
Bonferroni-corrected across 23 models tried	p=0.0044

Keeps 98%+ of its profit under realistic trading costs (10-40bps). Spans 16 companies, including 4 banks absent from training.

So, does Polymarket respond to news shocks?

Yes, and the picture sharpens as we look closer. Polymarket's own data shows a reaction (Result 1); every model we tested lands close to guessing the average (Results 2-4).

Checked against the real world, Polymarket's reaction tracks the real stock's direction (Result 5). For price-target contracts, that relationship strengthens into a tested, tradeable signal (Result 6).

Scoping decisions

- **market_cap_ladder** (4 markets): dropped, too small to mean anything
- **valuation_ladder** (110 markets): kept, flagged as below our 200-market floor, everything else about it clears comfortably
- **revenue_ladder**: given its own data split so it could be trained at all, rather than excluded

Limitations

- GDELT gives day-precision timestamps, a structural limit of this feed
- `resolved_at` and `parent_event_id` are each about 40% backfilled from Polymarket's API; fallbacks cover the rest
- The revenue-target result rests on 19 training examples
- The price-target trading signal (Result 6) spans 16 companies; transaction costs come from published research rather than Polymarket's measured spreads

Recommendation

- Lead the thesis with the matching result (Result 1) and the price-target trading signal (Result 6): the two clean, adversarially-tested contributions
- Present the consistency correction, the model comparison, and the external-benchmark proportionality check as negative or mixed results
- Keep the 200-market floor as the rule, with `valuation_ladder` as a documented, explicit exception
- Measuring Polymarket's actual historical bid-ask spread would sharpen Result 6's cost estimate — the highest-leverage next step